

DEVHEAL LABS AI

DH-ADOM

Technical specification and conformance requirements

Specification 1.0 | Layout revised 7 October 2026

Sai Narender Nuckala / DevHeal Labs AI

An engineering operating model for repository ownership, bounded delegation and evidence-backed handoffs.

Canonical publication: <https://devheallabs.com/dh-adom/>

Copyright (c) 2026 DevHeal Labs AI Pvt. Ltd. Publication prose and diagrams: Creative Commons Attribution 4.0 International (CC BY 4.0). <https://creativecommons.org/licenses/by/4.0/>

Embedded code examples: Apache-2.0. Names and logos excluded. Full scope: <https://devheallabs.com/dh-adom/licensing/> - licensing release 1.1.1.

DH-ADOM Technical Specification v1.0

Status: Normative

Version: 1.0.0

Date: 2026-10-06

1. Scope

DH-ADOM specifies a repository-level operating model for AI-native software engineering. It defines governance hierarchy, agent roles, domain ownership, bounded delegation, lifecycle integration, evidence, state, handoff, and conformance requirements.

2. Normative language

The terms MUST, MUST NOT, SHOULD, SHOULD NOT, MAY, and RECOMMENDED are normative.

3. Governance hierarchy

1. Engineering Constitution
2. AI-SDLC
3. ADLC
4. Root Agent
5. Feature Agent
6. Specialist Agent
7. Implementation

4. Agent roles

4.1 Root Agent

The Root Agent MUST own global orchestration and MUST NOT require itself to implement every feature.

4.2 Feature Agent

Every meaningful bounded domain SHOULD have one persistent Feature Agent. A Feature Agent MUST declare intent, ownership, capabilities, tools, permissions, autonomy, risks, state, and handoff.

4.3 Specialist Agent

A Specialist Agent **MUST** be created under a parent Feature Agent or explicitly authorized Root Agent. It **MUST** have a bounded purpose and task-scoped permissions.

5. Delegation Contract

Every delegation **MUST** identify parent, child, purpose, task, scope, permissions, budget, lifetime, evidence requirements, and acceptance criteria.

6. Delegation invariants

- Child scope **MUST** be a subset of authorized parent scope unless explicitly widened through governance.
- Delegation depth **MUST** be bounded.
- Parallel child execution **MUST** be bounded.
- Child permissions **MUST NOT** exceed delegated authority.
- Completion **MUST** return evidence and handoff.
- Uncontrolled recursive spawning **MUST** be blocked.

7. Feature ownership

Feature Agent ownership is an engineering boundary. Cross-feature changes require coordination through the Root Agent and **MUST** be traceable.

8. Identity and contract

Agent identity is distinct from model identity. Each agent **MUST** have an identity record and **SHOULD** have a machine-readable passport and contract.

9. AI-SDLC integration

All material work **MUST** map to intent, requirements, architecture, implementation, validation, evidence, release, and feedback.

10. ADLC integration

All agents **MUST** follow an agent lifecycle that includes intent, contract, identity, capabilities, permissions, autonomy, validation, promotion, monitoring, revalidation, and retirement.

11. Evidence

Meaningful tasks **MUST** emit auditable evidence of implementation and validation.

12. Handoff

A completed task **MUST** update state and create a machine-readable handoff sufficient for an independent AI session to continue work.

13. Security

Agents **MUST** follow least privilege. Production actions **MUST** be policy-controlled. Permission or autonomy escalation **MUST NOT** be self-authorized.

14. Conformance

An implementation conforms to DH-ADOM v1.0 only when all mandatory requirements in ``CONFORMANCE-REQUIREMENTS.md`` pass or have a formally accepted exception.

DH-ADOM v1.0 Conformance Requirements

An implementation **MUST** satisfy all **CRITICAL** requirements and provide evidence for each requirement.

ID	Requirement	Class
CR-001	Engineering Constitution exists and is authoritative	Critical
CR-002	Root Agent exists and is identifiable	Critical
CR-003	Feature Agent ownership exists for meaningful domains	Critical
CR-004	Specialist delegation is bounded	Critical
CR-005	Delegation contracts are auditable	Critical
CR-006	Root does not silently bypass Feature Agent ownership	Critical

ID	Requirement	Class
CR-007	AI-SDLC state is represented	High
CR-008	ADLC state is represented for agents	High
CR-009	Agent identity is separate from model identity	High
CR-010	Agent permissions are explicit	Critical
CR-011	Autonomy/delegation budgets are enforced	Critical
CR-012	Agent creation is bounded	Critical
CR-013	State and handoff are persisted	High
CR-014	Requirement-to-evidence traceability exists	High
CR-015	Audit trail records delegation and material actions	High
CR-016	Cross-feature changes are controlled	Critical
CR-017	Failure / timeout handling exists	High
CR-018	Fresh-session resumability is demonstrated	High
CR-019	No claimed completion without validation evidence	Critical
CR-020	Conformance results are reproducible	High